

AIDAN SCANNELL

Machine Learning Researcher

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"Machine Learning Researcher with 6+ years of experience developing and implementing machine learning and sequential decision-making algorithms, with a focus on learning world models and leveraging them for downstream decision-making. Proven ability to translate research ideas into high-performing implementations on multi-GPU systems, including generative AI approaches such as Transformers. Experienced in leading research teams, driving collaborations, and presenting work at top-tier conferences."

SKILLS

Python

C++

PyTorch

JAX

Hydra

Weights & Biases

Docker

SLURM

Git

LaTeX

ROS

Distributed ML

AMP

World models

Reinforcement learning

Generative modelling

Embodied AI

EXPERIENCE

Research Associate with Prof. Amos Storkey & Prof. Peter Bell

University of Edinburgh | Bayesian and Neural Systems group

 Jan 2025 – Ongoing

 Edinburgh, UK

- Won the 1X World Model Challenge and presented at ICCV 2025 workshop "Learning to See: Advancing Spatial Understanding for Embodied Intelligence".
- Developed Transformer- and diffusion-based world models for long-horizon and controllable generation.
- Bridged algorithmic design and scalable implementation, achieving 3× throughput in multi-GPU training pipelines.
- Presented invited talks on world models and scalable generative training at academic and industry venues.

World models

Transformers

Generative modelling

Scalable training

Multi-GPU systems

Embodied AI

Postdoctoral Researcher with Prof. Joni Pajarinen & Prof. Arno Solin

Aalto University | Finnish Center for Artificial Intelligence (FCAI)

 July 2022 – December 2024

 Helsinki, Finland

Received four-year funding to work jointly in the Robot Learning Lab and Machine Learning Group.

- Led cross-disciplinary collaborations with multiple PIs and PhD students to develop a generative world model for both decision-time and background planning; published at ICLR 2025.
- Co-led research on Bayesian deep learning for uncertainty-aware world models to guide exploration in model-based RL; published at ICLR 2024.
- Supervised MSc and PhD research on robotics foundation models, RL, and generative modeling for decision-making.
- Led FCAI's "Long-term Decision Making and Transfer Between Tasks" research team, defining technical strategy and mentoring early-career researchers.
- Scaled and optimized training on multi-GPU HPC clusters using PyTorch for world model and RL experiments.
- Maintained research codebases with documentation (Sphinx), testing, and open-source collaboration.

World models

Generative modelling

Reinforcement learning

Scalable training

Robotics

Leadership

Co-lecturer @ **Aalto University**

 Sept 2022 – December 2024

 Helsinki, Finland

- Co-lecturer for courses on *Reinforcement Learning* and *Advanced Gaussian Processes*, covering both theoretical foundations and practical implementation.
- Delivered lectures and tutorials, contributing to course design and receiving strong student feedback.

Teaching

Communication

Mentorship

Public speaking

Reinforcement learning

PhD Researcher (Supervisors: Prof. Arthur Richards & Prof. Carl Henrik Ek)

CDT in Future Autonomous and Robotic Systems, University of Bristol/Bristol Robotics Laboratory

📅 Sept 2018 – May 2022

📍 Bristol, UK

Awarded a four-year PhD scholarship including a taught MRes year.

- Combined methods from machine learning, stochastic geometry, and reinforcement learning to control quadcopters in uncertain real-world environments.
- Deployed learning-based control algorithms on physical robots using TensorFlow, JAX, and ROS.
- Presented findings at AISTATS and ICRA, bridging ML theory and robotic control.
- Teaching assistant for *Machine Learning*, *Robotic Systems*, and *Intelligent Information Systems* courses.

Machine learning

Reinforcement learning

Uncertainty quantification

Robotics

Stochastic geometry

👤 AWARDS

- Won both tracks of the 1X World Model Challenge and presented findings at ICCV 2025 Workshop.
- Awarded >300k GPU hours by LUMI Supercomputer for “Sample-efficient Large-scale Reinforcement Learning”.
- Awarded Postdoctoral Networking Tour in Artificial Intelligence (Postdoc-NeT-AI) fellowship by the German Academic Exchange Service (DAAD) for funded research visit to Germany.
- Received four-year postdoctoral funding from Finnish Center of Artificial Intelligence to work on ‘Model-based Reinforcement Learning under Uncertainty’.
- Awarded a four-year PhD scholarship at University of Bristol with the FARSCOPE Center for Doctoral Training.

INVITED TALKS

- Generative World Modelling for Humanoids, **ICCV 2025 Workshop on Learning to See: Advancing Spatial Understanding for Embodied Intelligence**, 19th October 2025, Hawaii, USA
- Discrete Codebook World Models, **Huawei-University of Edinburgh Joint Lab Workshop 2025**, 2nd April 2025, Xi'an, China
- Sample-Efficient RL with Implicitly Quantized Representations, **Nordic AI Meet + AI Day 2024**, 20th Oct 2024, Helsinki, Finland
- iQRL - Implicitly Quantized Representations for Sample-efficient RL, **International Workshop of Intelligent Autonomous Learning Systems**, 23 July 2024, Darmstädter Haus, Austria
- Model-based RL, **Cambridge Ellis Unit Summer School on Probabilistic Machine Learning 2024**, 17th July 2024
- Sequential Decision Making - Bayesian Optimization & Model-based RL, **Advanced course on Gaussian processes @ Aalto University**, 8 April 2024
- (Function-space) Laplace Approximation for BNNs, **National Science Foundation (NSF) Safe RL Team**, 3 Oct 2023
- Neural Networks as Sparse Gaussian processes for Sequential Learning, **International Workshop of Intelligent Autonomous Learning Systems**, 15 Aug 2023, Darmstädter Haus, Austria
- Model-based RL under uncertainty: the importance of knowing what you don't know, **RL course @ Aalto University**, Nov. 2023
- Sequential Decision Making, **Advanced course on Gaussian processes @ Aalto University**, 4 April 2023
- Model based RL under uncertainty, **ML at the Cambridge Computer Lab (ML@CL)**, 23 Feb 2023, University of Cambridge
- Synergising Bayesian Inference and Probabilistic Geometries for Robotic Control, **Cognitive Systems Group @ Technical University of Denmark (DTU)**, 18 March 2021

EDUCATION

PhD in Robotics and Autonomous Systems

University of Bristol

📅 Sept 2018 - June 2022

PhD Thesis:

- 📖 Bayesian Learning for Control in Multimodal Dynamical Systems

Taught MRes Year:

- First class honours (GPA=4.0/4.0)

Summer Schools:

- Machine Learning Summer School 2019
- Gaussian Process and Uncertainty Quantification Summer School 2019

MEng in Mechanical Engineering

University of Bristol | First Class Honours (GPA=4.0/4.0)

📅 Sept 2012 – June 2016

- Graduated in top 10% of cohort

REFERENCES

Prof. Arno Solin

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Prof. Joni Pajarinen

@ Aalto University

✉ joni.pajarinen@aalto.fi

Prof. Carl Henrik Ek

@ University of Cambridge

✉ che29@cam.ac.uk

Prof. Arthur Richards

@ University of Bristol

✉ arthur.richards@bristol.ac.uk

REVIEWING

- Area chair for European Workshop on Reinforcement Learning (EWRL)
- International Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- International Conference on Machine Learning (ICML)
- (Senior Reviewer) Reinforcement Learning Conference (RLC)
- IEEE Transaction on Pattern Analysis and Machine Intelligence (TPAMI)
- Annual Conference on Learning for Dynamics and Control (L4DC)
- International Conference on Artificial Intelligence and Statistics (AISTATS)
- Conference on Robot Learning (CoRL)
- International Conference on Robotics and Automation (ICRA)
- IEEE Robotics and Automation Letters (RA-L)

👤 PUBLICATIONS

👤 Publications

- Riccardo Mereu*, Aidan Scannell*, Yuxin Hou, Yi Zhao, Aditya Jitta, Antonio Dominguez, Luigi Acerbi, Amos Storkey, and Pauland Chang (2025). **Generative World Modelling for Humanoids: 1X World Model Challenge Technical Report**. In: *arxiv preprint arXiv:2510.07092*. * Equal contribution.
- Aidan Scannell, Mohammadreza Nakhaei, Kalle Kujanpää, Yi Zhao, Kevin Luck, Arno Solin, and Joni Pajarinen (2025). **Discrete Codebook World Models for Continuous Control**. In: *The Thirteenth International Conference on Learning Representations*.
- Yi Zhao, Aidan Scannell, Yuxin Hou, Tianyu Cui, Le Chen, Arno Solin, Juho Kannala, and Joni Pajarinen (2025). **Generalist World Model Pre-Training for Efficient Reinforcement Learning**. In: *ICLR 2025 Workshop on World Models: Understanding, Modelling and Scaling*.
- Yi Zhao, Aidan Scannell, Wenshuai Zhao, Yuxin Hou, Tianyu Cui, Le Chen, Arno Solin, Juho Kannala, and Joni Pajarinen (2025). **Efficient Reinforcement Learning by Guiding Generalist World Models with Non-Curated Data**. In: *arXiv preprint arXiv:2502.19544*.
- Mohammadreza Nakhaei, Aidan Scannell, and Joni Pajarinen (2024a). **Entropy Regularized Task Representation Learning for Offline Meta-Reinforcement Learning**. In: *AAAI Conference on Artificial Intelligence (AAAI)*.
- – (June 2024b). **Residual Learning and Context Encoding for Adaptive Offline-to-Online Reinforcement Learning**. In: *Proceedings of The 6th Annual Conference on Learning for Dynamics and Control*.
- Aidan Scannell, Kalle Kujanpää, Yi Zhao, Mohammadreza Nakhaei, Arno Solin, and Joni Pajarinen (2024a). **iQRL - Implicitly Quantized Representations for Sample-efficient Reinforcement Learning**. In: *arXiv preprint arXiv:2406.02696*.
- – (July 2024b). **Quantized Representations Prevent Dimensional Collapse in Self-predictive RL**. in: *ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET 2024)*.
- Aidan Scannell, Riccardo Mereu, Paul Chang, Ella Tami, Joni Pajarinen, and Arno Solin (May 2024). **Function-space Parameterization of Neural Networks for Sequential Learning**. In: *The Twelfth International Conference on Learning Representations (ICLR)*.
- Aidan Scannell, Carl Henrik Ek, and Arthur Richards (Apr. 2023). **Mode-constrained Model-based Reinforcement Learning via Gaussian Processes**. In: *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics (AISTATS)*.
- Aidan Scannell, Riccardo Mereu, Paul Chang, Ella Tami, Joni Pajarinen, and Arno Solin (July 2023). **Sparse Function-space Representation of Neural Networks**. In: *ICML 2023 Workshop on Duality Principles for Modern Machine Learning*.
- Aidan Scannell (2022). "Bayesian Learning for Control in Multimodal Dynamical Systems". PhD thesis. University of Bristol.
- Aidan Scannell, Carl Henrik Ek, and Arthur Richards (June 2021). **Trajectory Optimisation in Learned Multimodal Dynamical Systems Via Latent-ODE Collocation**. In: *2021 IEEE International Conference on Robotics and Automation (ICRA)*.